

Language Models as Reads over Atom Spaces: A Unified Measure-Theoretic Formulation of Context, Retrieval, and Parameters

Farshid Mossaiby^{a,*}

^aDepartment of Civil Engineering, University of Isfahan, 81744-73441, Isfahan, Iran

Abstract

We express finite attention, external retrieval, and expert-output routing through a common read operator: an integral of a query-conditioned measure against a value map over a valued atom space $(\Omega, \mathcal{F}, \nu, \nu)$. This is an output-space abstraction rather than an assertion that the mechanisms have identical internal computations. We give a sufficient condition for a newly inserted atom to enter an exact top- k set, prove two truncation bounds, and derive an exact-tail rule for selecting the minimum k under a requested error budget. On CounterFact-500, frozen MiniLM exact retrieval obtains 99.2% rewrite and 91.0% paraphrase top-1 recall, but a fixed 0.60 activation threshold retains only 38.2% of paraphrases and falsely activates on 9.6% of neighborhood prompts. Across five seeds, surface-matched context/corpus/parameter route accuracies are $100.0 \pm 0.0\%$, $82.4 \pm 0.9\%$, and $100.0 \pm 0.0\%$. Matched negative atoms cancel paired signed-read contributions; a perturbation bound and 3000-trial stress test quantify degradation away from exact matching. A ten-fact GPT-2 mechanism study obtains 10/10 canonical and 9/10 paraphrase first-token corrections while ten unrelated controls bypass intervention. These studies do not establish superiority over parametric editing, large-scale ANN performance, certified unlearning, or general factuality improvement.

Keywords: Language models, Atom spaces, Measure theory, External memory, Vector retrieval, Expert routing

*Corresponding author.

Email address: mossaiby@eng.ui.ac.ir (Farshid Mossaiby)

1. Introduction

Modern autoregressive language models conflate two fundamentally distinct computational roles into dense parameter matrices: general linguistic syntax and factual world knowledge [1, 2]. While gradient descent is well-suited for acquiring syntactic abstractions, semantic parsing, and structural reasoning, compiling static factual assertions directly into dense weights leads to severe practical bottlenecks. Factual knowledge is dynamic, highly localized, and subject to continuous obsolescence and legal erasure requirements (e.g., the GDPR “Right to be Forgotten”). When knowledge is frozen within billions of dense parameters, modifying or updating a single fact necessitates expensive retraining or fragile post-hoc weight editing [3, 4].

To circumvent the rigidity of dense parameter compilation, the community has developed disparate engineering solutions:

- *In-context reasoning* stores temporary information in the key-value cache of the sequence window [1].
- *Retrieval-augmented generation* (RAG) and nearest-neighbor language models (k NN-LM) query external vector databases of textual records [5, 6, 7].
- *Modular computation* mechanisms, such as Mixture-of-Experts (MoE) and parameter-efficient adapters (LoRA), route queries across discrete banks of specialized sub-networks [8, 9, 10].

Although these techniques share an underlying reliance on key-value similarity matching [11, 12], they have evolved as disconnected architectural paradigms with distinct mathematical formulations, ad-hoc routing heuristics, and poorly understood stability limits.

In this work, we propose a common measure-theoretic notation for context, external retrieval, and parameter-expert output routing. Attention can be written as an integral of a query-conditioned probability measure against a vector-valued map. Corpus values are fixed records, whereas parameter-expert values may depend on the query; the abstraction identifies their shared weighted-read structure without claiming identical internal computation.

By establishing this equivalence, we make the following contributions:

1. **A Common Mathematical Abstraction:** We formalize the read operator $R(x; \Omega)$ and define an integrated tri-space architecture that dynamically arbitrates between context (Ω_{ctx}), corpus retrieval (Ω_{corpus}), and parameter experts (Ω_{params}).

- 36 **2. A Sufficient Retrieval Condition:** We derive a Lipschitz-margin condition
 37 sufficient for a new atom to enter an exact top- k set; prediction success requires
 38 additional value and output-margin assumptions.
- 39 **3. The Quadrature Truncation Bound Theorem:** We prove Theorem 1, show-
 40 ing exponential decay of exact ranked top- k truncation error when temperature
 41 is calibrated relative to the retrieval gap. ANN candidate quality is a separate
 42 approximation not analyzed by the theorem.
- 43 **4. Signed External-Memory Suppression:** Extending the read to signed ref-
 44 erence measures ($v = v^+ - v^-$) permits matched negative atoms to cancel
 45 external-memory contributions without retraining; this does not claim removal
 46 from pretrained parameters.
- 47 **5. Controlled Mechanism Studies:** Synthetic and CounterFact experiments
 48 examine insertion, adaptive exact truncation, HNSW candidates, cumulative
 49 memory, routing, signed-read suppression, and GPT-2 steering. Trainable
 50 synthetic studies use five seeds. The implementation and precomputed logs
 51 are available at <https://github.com/mossaiby/RoAS>.

52 **2. From a Discrete Function to an Integral**

53 *2.1. The Token-Level Function*

54 A language model is conventionally defined as a conditional probability dis-
 55 tribution over a finite vocabulary V of size d , given an input prefix of variable
 56 length. To remove indexing dependencies over variable sequence lengths, we fix
 57 a maximum context capacity c and introduce a null padding symbol $\emptyset$, yielding
 58 an augmented vocabulary $\bar{V} = V \cup \{\emptyset\}$. Every input sequence is represented as
 59 a fixed-dimensional tuple

$$x = (x_1, \dots, x_c) \in \bar{V}^c,$$

60 where unused positions are padded with $\emptyset$. A language model is a mapping

$$p : \bar{V}^c \rightarrow \Delta^{d-1}, \quad \Delta^{d-1} = \left\{ z \in \mathbb{R}^d : z_i \geq 0, \sum_{i=1}^d z_i = 1 \right\}.$$

2.2. Attention as an Integral

Let $\phi : \bar{V} \rightarrow \mathbb{R}^e$ denote a token embedding map such that $\phi(\emptyset) = 0$. We extend the embedded sequence to a step function defined over the continuous interval $[0, c]$:

$$f_x(t) = \phi(x_{\lceil t \rceil}), \quad t \in (0, c],$$

with $f_x(0) = \phi(x_1)$. Let $\alpha_x : [0, c] \rightarrow \mathbb{R}_{\geq 0}$ be a context-dependent attention weight density satisfying the normalization condition $\int_0^c \alpha_x(t) dt = 1$, with $\alpha_x(t) = 0$ whenever $x_{\lceil t \rceil} = \emptyset$. We define the aggregate context representation vector as

$$z(x) = \int_0^c \alpha_x(t) f_x(t) dt \in \mathbb{R}^e,$$

and the corresponding token prediction distribution as

$$p(x) = \text{softmax}(Wz(x) + b).$$

Because f_x is piecewise constant over unit intervals, the continuous integral evaluates algebraically to the standard attention weighted sum [1, 13]:

$$\int_0^c \alpha_x(t) f_x(t) dt = \sum_{i=1}^c \left(\int_{i-1}^i \alpha_x(t) dt \right) \phi(x_i) = \sum_{i=1}^c \alpha_i \phi(x_i).$$

While this establishes algebraic equivalence for standard sequence self-attention, the continuous formulation suggests an immediate generalization: replacing the bounded token coordinate domain $[0, c]$ with an arbitrary, potentially mutable *atom space*.

3. Atom Spaces: A Common Object for Context, Retrieval, and Parameters

Definition 1 (Valued Atom Space). A valued atom space is a tuple $(\Omega, \mathcal{F}, \nu, v)$, where $(\Omega, \mathcal{F})$ is a measurable space of elements designated as atoms, ν is a σ -finite reference measure, and $v : \Omega \rightarrow \mathbb{R}^e$ is a measurable value map into a common output representation space.

Definition 2 (Query-Conditioned Kernel and Read Operator). Given a query $x \in \mathcal{X}$, let $K_x : \Omega \rightarrow \mathbb{R}_{\geq 0}$ be measurable and suppose

$$0 < Z_x := \int_{\Omega} K_x(\omega) d\nu(\omega) < \infty.$$

Table 1: Three structural realizations of the unified read operator $R(x; \Omega)$.

Instance	Atom Space Ω	Atom ω	Value Map $v(\omega)$	Operational Equivalent
Context	$\{1, \dots, c\}$	Token index	Embedding $\phi(x_\omega)$	In-context self-attention [1]
Corpus	$\mathcal{D}$ (mutable)	Factual record	Encoded fact vector	Dense retrieval / k NN-LM [6]
Parameters	Θ (expert bank)	Modular expert	Expert output $g_\omega(x) \in \mathbb{R}^e$	MoE-style output routing [8, 9]

82 Then K_x induces a query-conditioned probability measure μ_x on $(\Omega, \mathcal{F})$ via the
 83 Radon–Nikodym derivative:

$$d\mu_x(\omega) = \frac{K_x(\omega)}{Z_x} dv(\omega).$$

84 If $\int_{\Omega} \|v(\omega)\|_2 d\mu_x(\omega) < \infty$, the read conditioned on x is the Bochner integral:

$$R(x; \Omega) = \int_{\Omega} v(\omega) d\mu_x(\omega) \in \mathbb{R}^e.$$

85 As shown in Table 1, in-context attention, corpus retrieval, and parameter rout-
 86 ing are all structural realizations of the read operator $R(x; \Omega)$, differing only in their
 87 domain of integration.

88 3.1. The Unified Architecture

89 We formulate an integrated architecture where context, external corpus mem-
 90 ory, and parameter routing feed into a unified representation:

$$z(x) = \lambda_{\text{ctx}}(x)R(x; \Omega_{\text{ctx}}) + \lambda_{\text{corpus}}(x)R(x; \Omega_{\text{corpus}}) + \lambda_{\text{params}}(x)R(x; \Omega_{\text{params}}),$$

91

$$p(x) = \text{softmax}(W_0 z(x) + b_0),$$

92 where $\lambda(x) \in \Delta^2$ is an input-dependent gating distribution across the three spaces,
 93 and $W_0 \in \mathbb{R}^{d \times e}$, $b_0 \in \mathbb{R}^d$ represent the linguistic projection head. For param-
 94 eter experts the value is query-dependent, $v_x(\omega) = g_\omega(x)$; the read is therefore an
 95 output-space mixture, not an average of parameter tensors. Corpus-memory up-
 96 dates can be implemented by inserting, deleting, or reweighting records without
 97 changing dense model weights, whereas changing a parameter expert generally
 98 requires parameter optimization.

99 3.2. System, Calibration, and Tying Principles

- 100 • **Scale Mismatch and Norm Calibration:** The cardinalities of the respective
 101 spaces vary across multiple orders of magnitude ($|\Omega_{\text{ctx}}| \sim 10^3$, $|\Omega_{\text{params}}| \sim$
 102 10^1 , $|\Omega_{\text{corpus}}| \sim 10^9$). Normalizing a joint softmax across the disjoint union
 103 $\Omega_{\text{ctx}} \sqcup \Omega_{\text{corpus}} \sqcup \Omega_{\text{params}}$ causes the largest space to artificially dominate proba-
 104 bility mass. Separate intra-space measure normalization combined with cali-
 105 brated convex gating $\lambda(x)$ is required. Furthermore, key representations across
 106 distinct spaces must share identical geometric scales (specifically, unit L_2 -
 107 normalization) to prevent dot-product magnitudes from systematically skewing
 108 the router toward higher-norm representations.
- 109 • **Tied Value-to-Token Embeddings:** In token generation tasks, the read vector
 110 $z_{\text{corpus}}(x)$ should project predictably onto target vocabulary tokens. Letting $E \in$
 111 $\mathbb{R}^{d \times e}$ denote the vocabulary embedding matrix, we use logits $Ez + b$ and values
 112 $v(\omega) = E[y_\omega] / \|E[y_\omega]\|$. Isolating atom ω^* therefore aligns $z(x)$ with the target
 113 row. A unique target argmax additionally requires that this alignment exceed
 114 the alignment with every competing output row after accounting for bias.

115 3.3. Signed Atom Spaces and the Hahn–Jordan Decomposition

116 To support cancellation of external-memory contributions without index re-
 117 construction, we extend the reference measure ν from a standard non-negative
 118 measure to a *signed measure*. By the Hahn–Jordan Decomposition Theorem,
 119 any signed reference measure uniquely decomposes into mutually singular non-
 120 negative measures:

$$\nu = \nu^+ - \nu^-, \quad \nu^+ \perp \nu^-,$$

121 where ν^+ governs positive constructive atoms and ν^- governs negative “anti-
 122 atoms”. Writing $|\nu| = \nu^+ + \nu^-$, assume $0 < Z_x^\pm = \int_\Omega K_x d|\nu| < \infty$. The
 123 normalized signed query measure is

$$d\mu_x^\pm(\omega) = \frac{K_x(\omega)}{Z_x^\pm} (d\nu^+(\omega) - d\nu^-(\omega)).$$

124 This is not a probability measure: its total variation is normalized, while its signed
 125 total mass need not equal one. If an anti-atom ω^- has the same key and value as
 126 a positive atom ω^* and equal positive mass under ν^- and ν^+ , respectively, their
 127 contributions cancel exactly inside the signed external read.

128 4. Mathematical Conditions for Post-Hoc Knowledge Editing

129 Let the query-key kernel be defined as a scaled inner product:

$$K_x(\omega) = \exp \left(\frac{\langle \phi_q(x), \phi_k(\omega) \rangle}{\tau} \right),$$

130 where $\phi_q : \mathcal{X} \rightarrow \mathbb{R}^m$ and $\phi_k : \Omega \rightarrow \mathbb{R}^m$ are query and key encoders, and $\tau > 0$
 131 is a temperature parameter. During base training, ϕ_q and ϕ_k are optimized against
 132 an initial atom pool $\Omega_{\text{train}} \subset \Omega$. Post-hoc editing requires that inserting a novel,
 133 unseen atom $\omega^* \notin \Omega_{\text{train}}$ at test time reliably captures the measure $\mu_x(\omega^*)$ for an
 134 associated query x^* , without modifying the parameters of ϕ_q or ϕ_k .

135 4.1. Kernel Locality

136 **Definition 3** (Metric Locality). *The key encoder ϕ_k is L -Lipschitz continuous with*
 137 *respect to a semantic pseudo-metric d_Ω on Ω if*

$$\|\phi_k(\omega) - \phi_k(\omega')\|_2 \leq L d_\Omega(\omega, \omega') \quad \forall \omega, \omega' \in \Omega.$$

138 **Proposition 1** (Score Stability on Unseen Atoms). *Let ϕ_k be L -Lipschitz with*
 139 *respect to d_Ω . For any training anchor $\omega_0 \in \Omega_{\text{train}}$ and any newly inserted atom*
 140 *$\omega^* \in \Omega$, the inner-product similarity score satisfies:*

$$\left| \langle \phi_q(x), \phi_k(\omega^*) \rangle - \langle \phi_q(x), \phi_k(\omega_0) \rangle \right| \leq \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0).$$

141 *Proof.* By the Cauchy–Schwarz inequality and the Lipschitz condition:

$$\begin{aligned} \left| \langle \phi_q(x), \phi_k(\omega^*) - \phi_k(\omega_0) \rangle \right| &\leq \|\phi_q(x)\|_2 \|\phi_k(\omega^*) - \phi_k(\omega_0)\|_2 \\ &\leq \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0). \end{aligned}$$

□

142 4.2. The Retrieval Margin Condition

143 Let $\tau_k(x) = \max_{\omega \in \Omega_{\text{train}}}^{(k)} \langle \phi_q(x), \phi_k(\omega) \rangle$ denote the k -th highest score among ex-
 144 isting training atoms. For the novel atom ω^* to enter the top- k retrieved set under
 145 query x , it must satisfy $\langle \phi_q(x), \phi_k(\omega^*) \rangle > \tau_k(x)$. Combining this with Proposi-
 146 tion 1 yields a checkable sufficient condition:

$$\langle \phi_q(x), \phi_k(\omega_0) \rangle - \|\phi_q(x)\|_2 L d_\Omega(\omega^*, \omega_0) > \tau_k(x), \quad (1)$$

147 where ω_0 is a training anchor semantically proximal to ω^* . Equation (1) is a suf-
 148 ficient condition only for top- k inclusion. Successful prediction additionally de-
 149 pends on retrieved probability mass, the value representation, the output margin,
 150 and behavior on control queries.

151 **Remark 1** (The Density–Lipschitz Trade-off). Equation (1) shows that top- k in-
 152 clusion depends on both local geometric proximity $d_\Omega(\omega^*, \omega_0)$ and the magnitude
 153 of L . Denser training pools may reduce the distance to a useful anchor, but may
 154 also alter encoder norms and competing score margins. Whether density system-
 155 atically changes the Lipschitz constant is an empirical hypothesis rather than a
 156 consequence of the proposition.

157 4.3. Anti-Atoms and Exact External-Read Cancellation

158 **Proposition 2** (Matched-Pair Cancellation). Let an atom $\omega^* \in \Omega_{\text{corpus}}$ encode
 159 a target fact with tied token value $v^* = E[y^*]/\|E[y^*]\|$. If an anti-atom ω^- is
 160 inserted into the signed atom space with identical key $\phi_k(\omega^-) = \phi_k(\omega^*)$, identical
 161 value $v^- = v^*$, and equal positive masses $v^-(\{\omega^-\}) = v^+(\{\omega^*\})$, then for any
 162 query x :

$$R^\pm(x; \{\omega^*, \omega^-\}) = \mathbf{0}.$$

163 *Proof.* Under the signed measure integral of Section 3.3, the contribution of the
 164 pair evaluates to:

$$\int_{\{\omega^*, \omega^-\}} v(\omega) d\mu_x(\omega) = \frac{K_x(\omega^*)v^* - K_x(\omega^-)v^*}{Z} = \frac{K_x(\omega^*)(v^* - v^*)}{Z} = \mathbf{0}.$$

165 This establishes cancellation of the matched pair’s contribution to the signed ex-
 166 ternal read. $\square$

167 Proposition 2 does not imply that the complete model representation is zero:
 168 background atoms, context, parameter experts, residual model states, and output
 169 bias may remain. Uniform output follows only in the special case where every other
 170 contribution and the output bias are exactly zero. The positive record and anti-atom
 171 remain stored, and the proposition does not imply removal of information encoded
 172 in pretrained model parameters.

173 **Proposition 3** (Approximate Pair Cancellation). Let $a = K_x(\omega^*)v^+(\{\omega^*\})$, $b =$
 174 $K_x(\omega^-)v^-(\{\omega^-\})$, and let $Z_x^\pm > 0$ be the total-variation normalization of the com-
 175 plete signed read. For a positive value v and anti-atom value $\tilde{v}$, the residual pair
 176 contribution satisfies

$$\left\| \frac{av - b\tilde{v}}{Z_x^\pm} \right\|_2 \leq \frac{|a - b| \|v\|_2 + b \|v - \tilde{v}\|_2}{Z_x^\pm}.$$

177 *Proof.* Write $av - b\tilde{v} = (a - b)v + b(v - \tilde{v})$ and apply the triangle inequality, then
 178 divide by $Z_x^\pm$. $\square$

Thus exact cancellation is sensitive to key-induced kernel mismatch, unequal signed mass, and value mismatch. Query paraphrases affect cancellation only through their effect on the two kernel weights; identical keys still cancel for every query.

5. Computational Tractability and the Quadrature Truncation Bound

Evaluating $R(x; \Omega) = \int_{\Omega} v(\omega) d\mu_x(\omega)$ over a large corpus is expensive if performed densely. We analyze numerical truncation to the exact ranked top- k elements:

$$\hat{R}_k(x; \Omega) = \sum_{i=1}^k v(\omega_{(i)}) \frac{K_x(\omega_{(i)})}{\sum_{j=1}^k K_x(\omega_{(j)})}.$$

The reference implementation computes all N scores and therefore costs $\mathcal{O}(Ne)$ before the top- k read. In deployment, an ANN index may reduce candidate retrieval cost, after which read aggregation costs

$$\mathcal{O}(c \cdot e) + \mathcal{O}(k_{\text{corpus}} \cdot e) + \mathcal{O}(k_{\text{params}} \cdot e).$$

We now formalize the conditions under which this numerical truncation faithfully approximates the continuous integral.

Let $N = |\Omega|$, and let $V_{\max} := \sup_{\omega \in \Omega} \|v(\omega)\|_2 < \infty$. For query x , let $s_i := \langle \phi_q(x), \phi_k(\omega_{(i)}) \rangle$, and let indices be ordered such that $s_{(1)} \geq s_{(2)} \geq \dots \geq s_{(N)}$. Define the normalization terms $Z := \sum_{j=1}^N e^{s_{(j)}/\tau}$ and $Z_k := \sum_{j=1}^k e^{s_{(j)}/\tau}$. We define the *retrieval spectral gap* at rank k as:

$$\Delta_k(x) := s_{(1)} - s_{(k+1)} \geq 0.$$

Theorem 1 (Quadrature Truncation Error Bound). *For any query x and any truncation depth $k \in \{1, \dots, N-1\}$, the Euclidean approximation error of the top- k quadrature satisfies:*

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq 2V_{\max} \cdot (N - k) \cdot \exp\left(-\frac{\Delta_k(x)}{\tau}\right).$$

An alternative bound is:

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq 2V_{\max} \cdot \left(\frac{N - k}{k}\right) \cdot \exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right).$$

200 *Proof.* Decompose the difference between the full integral and the top- k trunca-
 201 tion:

$$\begin{aligned} R(x; \Omega) - \hat{R}_k(x; \Omega) &= \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} \left(\frac{1}{Z} - \frac{1}{Z_k} \right) + \sum_{i=k+1}^N v_{(i)} \frac{e^{s_{(i)}/\tau}}{Z} \\ &= \frac{Z_k - Z}{Z Z_k} \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} + \frac{1}{Z} \sum_{i=k+1}^N v_{(i)} e^{s_{(i)}/\tau}. \end{aligned}$$

202 Let $Z_{\text{tail}} := Z - Z_k = \sum_{i=k+1}^N e^{s_{(i)}/\tau}$. Notice that $\frac{1}{Z_k} \sum_{i=1}^k v_{(i)} e^{s_{(i)}/\tau} = \hat{R}_k(x; \Omega)$, and
 203 define the tail center of mass $R_{\text{tail}} := \frac{1}{Z_{\text{tail}}} \sum_{i=k+1}^N v_{(i)} e^{s_{(i)}/\tau}$. Substituting yields:

$$R(x; \Omega) - \hat{R}_k(x; \Omega) = \frac{Z_{\text{tail}}}{Z} (R_{\text{tail}} - \hat{R}_k(x; \Omega)).$$

204 Taking Euclidean norms and applying the triangle inequality:

$$\|R(x; \Omega) - \hat{R}_k(x; \Omega)\|_2 \leq \frac{Z_{\text{tail}}}{Z} (\|R_{\text{tail}}\|_2 + \|\hat{R}_k(x; \Omega)\|_2) \leq 2V_{\max} \frac{Z_{\text{tail}}}{Z}.$$

205 To bound the tail ratio $\frac{Z_{\text{tail}}}{Z}$, note that $Z \geq e^{s_{(1)}/\tau}$ and $Z_{\text{tail}} = \sum_{i=k+1}^N e^{s_{(i)}/\tau} \leq (N -$
 206 $k)e^{s_{(k+1)}/\tau}$:

$$\frac{Z_{\text{tail}}}{Z} \leq \frac{(N - k)e^{s_{(k+1)}/\tau}}{e^{s_{(1)}/\tau}} = (N - k) \exp\left(-\frac{s_{(1)} - s_{(k+1)}}{\tau}\right).$$

207 Substituting this into the norm inequality proves the first bound. Alternatively,
 208 $Z \geq k e^{s_{(k)}/\tau}$ while $Z_{\text{tail}} \leq (N - k)e^{s_{(k+1)}/\tau}$, giving

$$\frac{Z_{\text{tail}}}{Z} \leq \frac{N - k}{k} \exp\left(-\frac{s_{(k)} - s_{(k+1)}}{\tau}\right),$$

209 which proves the second bound. Neither bound uniformly dominates the other; the
 210 second is smaller when $s_{(1)} - s_{(k)} < \tau \log k$. $\square$

211 **Corollary 1** (Error-Budgeted Exact Truncation). *Let $\rho_k(x) = Z_{\text{tail}}/Z$ be the exact*
 212 *softmax mass outside the first k ranked atoms. For any requested $\varepsilon > 0$, define*

$$k_\varepsilon(x) = \min\{k \in \{1, \dots, N\} : 2V_{\max} \rho_k(x) \leq \varepsilon\}.$$

213 *Then $\|R(x; \Omega) - \hat{R}_{k_\varepsilon(x)}(x; \Omega)\|_2 \leq \varepsilon$.*

214 *Proof.* The proof of Theorem 1 first establishes $\|R - \hat{R}_k\|_2 \leq 2V_{\max}\rho_k$ before
 215 bounding ρ_k by score gaps. Substitution of the defining inequality for $k_\epsilon(x)$ gives
 216 the result. $\square$

217 Theorem 1 bounds truncation after the exact score ordering is known: when τ
 218 is small relative to $\Delta_k(x)$, the omitted tail decays exponentially. Corollary 1 turns
 219 the exact tail mass into a per-query stopping rule, but computing that certificate
 220 still requires all scores and therefore is not a sublinear retrieval algorithm. An
 221 ANN system introduces an additional candidate-recall error and requires separate
 222 empirical evaluation.

223 6. Empirical Evaluation

224 We examine inserted-record retrieval, adaptive exact top- k truncation and scal-
 225 ing, CounterFact retrieval with HNSW candidates, cumulative insertion, tri-space
 226 routing, signed-read suppression under perturbation, and GPT-2 steering.

227 6.1. Experimental Setup

228 *Corpus and Domain Design.* We construct a benchmark of 200 structured tech-
 229 nical facts distributed equally across four domains: *Astronomy & Astrophysics*,
 230 *Systems & Computing*, *Chemistry & Materials Science*, and *Molecular Biology &*
 231 *Genetics*. Each triplet (s, r, o) is mapped to a declarative record, a canonical query,
 232 and one mechanically generated paraphrase. The first 40 items per domain form
 233 the adapter-training pool; the remaining 10 items per domain are excluded from
 234 adapter training and used as post-hoc insertions. Their labels remain present in the
 235 fixed global output vocabulary, making this a transductive label-space evaluation
 236 rather than a fully unseen-label split.

237 *Encoder Architecture and Inductive Bias.* We employ a frozen sentence trans-
 238 former (sentence-transformers/all-MiniLM-L6-v2, $d_{\text{in}} = 384$) as the ge-
 239 ometric backbone. To prevent the metric collapse observed when training un-
 240 constrained projection heads from scratch on small corpora, we adopt a metric-
 241 preserving residual adapter:

$$\phi(x) = \text{Normalize}(x + 0.1 \cdot \text{MLP}(x)),$$

242 where the final layer of the MLP is initialized to zero. Values are tied to unit-
 243 normalized token embeddings: $v(\omega) = E[y_\omega] / \|E[y_\omega]\|$, with predictions eval-
 244 uated via the tied projection $p(y \mid x) \propto \langle z(x), E[y] \rangle / \tau$. All engines, bench-
 245 mark definitions, aggregation scripts, and plotting routines are available at [https:](https://github.com/mossaiby/RoAS)
 246 [/github.com/mossaiby/RoAS](https://github.com/mossaiby/RoAS).

Table 2: Inserted-record retrieval across 200 templated technical facts. Efficacy and generality denote target classification for a canonical query and one generated paraphrase. Specificity denotes unchanged argmax predictions on active training queries. The adapter spectral product is a diagnostic, not a Lipschitz certificate.

Metric	1 / topic	3 / topic	10 / topic	40 / topic
<i>Trained Residual Read</i>				
Efficacy (Target Recall on ω^*)	100.0%	100.0%	100.0%	100.0%
Generality (Paraphrase Query)	100.0%	100.0%	100.0%	100.0%
Specificity (Neighborhood Retention)	100.0%	100.0%	100.0%	100.0%
<i>Frozen MiniLM Exact 1-Nearest-Neighbor Baseline</i>				
Efficacy (Target Recall on ω^*)	100.0%	100.0%	100.0%	100.0%
Generality (Paraphrase Query)	100.0%	100.0%	100.0%	100.0%
Specificity (Argmax Retention)	100.0%	100.0%	100.0%	100.0%
<i>Representation and Metric Dynamics</i>				
Novel Atom Attention Mass $\mu_x(\omega^*)$	99.97%	99.97%	99.94%	99.75%
Adapter Linear Spectral Product	0.469	0.818	0.859	1.003

247 *Reproducibility Environment.* Committed artifacts were generated using Python
248 3.11.9, PyTorch 2.6.0+cu124, NumPy 2.4.6, sentence-transformers 5.7.0, trans-
249 formers 4.48.3, and torchvision 0.21.0 on an NVIDIA GeForce RTX 3050 Ti Lap-
250 top GPU (4 GB VRAM) with an AMD Ryzen 7 5800U host system and 15.4 GB
251 RAM. Package versions are pinned in `requirements.txt`. The routing sweep re-
252 ports seeds {40, 41, 42, 43, 44} and sample standard deviations; seed 42 is retained
253 as the canonical detailed run. Other experiments use deterministic seed 42.

254 6.2. Post-Hoc Knowledge Editing Results

255 Table 2 compares the trained read with frozen MiniLM exact 1-nearest-
256 neighbor retrieval across pool densities $\Omega_{\text{train}} \in \{1, 3, 10, 40\}$ items per domain.

257 Across the 40 insertion records and all density conditions, both methods
258 obtain 100% target classification for the canonical query and the single tem-
259 plated paraphrase, with no argmax changes on active training queries. The frozen
260 exact-retrieval baseline therefore matches the trained read: the benchmark does
261 not demonstrate that the residual adapter improves retrieval. At density 40, the
262 trained read’s maximum output-probability shift is nonzero (2.86×10^{-6}), showing
263 stability on the evaluated controls rather than zero representational distortion or

Table 3: Quadrature truncation error $\|R_{\text{exact}} - \hat{R}_k\|_2$ evaluated on context vectors across temperatures τ and retrieval cutoffs k . Under unit-normalized embeddings ($V_{\text{max}} = 1$), error is strictly bounded by 2.0 and exhibits exponential decay at sharp operational temperatures ($\tau = 0.05, k = 1 \Rightarrow 3.82 \times 10^{-6}$).

Temperature	Top-1	Top-3	Top-5	Top-10	Top-20
$\tau = 0.05$	3.82×10^{-6}	3.01×10^{-6}	2.62×10^{-6}	2.03×10^{-6}	1.41×10^{-6}
$\tau = 0.10$	0.0182	0.0170	0.0162	0.0145	0.0121
$\tau = 0.50$	0.965	0.700	0.558	0.384	0.245
$\tau = 1.00$	0.988	0.623	0.477	0.327	0.217

superiority to parametric editing; competitive baselines such as ROME, MEMIT, WISE, AlphaEdit, and generative RAG are not evaluated here [3, 4, 14, 15].

As training density scales from 1 to 40 atoms per domain, the product of the adapter’s linear-layer spectral norms increases from 0.469 to 1.003, while attention mass decreases from 99.97% to 99.75%. This product is a diagnostic of the MLP’s linear components, not a certified Lipschitz constant for the complete residual-normalization map ϕ ; directly estimating that constant remains future work.

6.3. Convergence of Exact Ranked Top-k Truncation

Table 3 presents the empirical quadrature truncation error $\|R_{\text{exact}}(x) - \hat{R}_k(x)\|_2 = \|z_{\text{exact}}(x) - z_k(x)\|_2$ across temperatures and cutoffs k on the largest atom pool (density 40, containing 160 active background atoms).

The numerical results illustrate the temperature dependence of exact truncation. We also compute both theorem bounds for every query. At density 40, $\tau = 0.10$, and $k = 1$, the maximum observed error is 0.0244 and the smaller point-wise bound covers 100% of queries with minimum slack 0.0923. This checks the implemented inequalities in this finite numerical setting rather than establishing bound tightness or ANN behavior:

- **Concentration at Low Temperatures ($\tau = 0.05$):** At the operational reading temperature, the mean error for $k = 1$ is 3.82×10^{-6} , decaying to 1.41×10^{-6} at $k = 20$. In this benchmark, the query measure is concentrated almost entirely on the nearest scored records.
- **Diffuse Measure Truncation at High Temperatures ($\tau \geq 0.50$):** When the kernel temperature is uncalibrated, probability mass spreads broadly across background atoms, producing truncation errors approaching 1.0. Exact top- k

Table 4: CounterFact-500 external-memory retrieval. “Activated correct” applies cosine threshold 0.60 after exact top-1 retrieval. Neighborhood activation is undesirable. HNSW candidate recall asks whether its ten candidates contain the exact top-1.

Evaluation	Metric	Result
Rewrite prompts	Exact top-1 / activated correct	99.2% / 99.2%
Paraphrase prompts	Exact top-1 / activated correct	91.0% / 38.2%
Neighborhood prompts	False activation / paired wrong-edit retrieval	9.6% / 2.4%
Adaptive read ($\epsilon = 0.05$)	Mean k / maximum error / coverage	246.0 / 0.0237 / 100%
Fixed read ($k = 10$)	Mean / maximum vector error	0.1769 / 0.4461
HNSW, 10 candidates	Exact top-1 candidate recall	99.96%

truncation is therefore faithful only when paired with sharp, metric-calibrated kernels.

Adaptive error budgets and scaling. We additionally evaluate Corollary 1 on 64 correlated synthetic queries with unit-normalized random keys and values in $\mathbb{R}^{128}$, for $N \in \{1000, 5000, 10000, 25000\}$. Across both temperatures and all three budgets $\epsilon \in \{0.01, 0.05, 0.10\}$, every measured read error satisfies its certificate (100% empirical coverage). The result also exposes conservatism: at $N = 25,000$ and $\epsilon = 0.05$, mean k_ϵ is 13,026 (52.1% of the corpus) for $\tau = 0.05$ and 21,467 (85.9%) for $\tau = 0.10$. Exact scoring, sorting, and reading take 54.6 and 18.2 ms, respectively, on the reported GPU run. Thus the rule provides a checkable accuracy budget but does not guarantee a small k for diffuse or weakly separated score distributions.

6.4. CounterFact External-Memory Retrieval and ANN Candidates

We evaluate the frozen MiniLM retrieval geometry on the first 500 CounterFact records introduced with ROME [3]. Each external-memory key is the benchmark rewrite prompt concatenated with its requested new target; evaluation uses 500 rewrite prompts, the first two paraphrases per record (1000 queries), and the first two neighborhood prompts (1000 queries). This is an external-memory retrieval experiment, not a parametric editing comparison with ROME or MEMIT.

Table 4 shows that pretrained embedding geometry retrieves most corresponding records, but the fixed threshold rejects a majority of otherwise-correct paraphrase matches. A FAISS HNSW index obtains 99.96% exact-top-1 candidate recall in this 500-record run, with 12.7 ms construction and 23.5 ms search for all 2500 queries. These timings are implementation- and hardware-specific, and the corpus is too small to establish large-scale ANN behavior.

Table 5: Cumulative insertion performance across 40 sequential records without teardown. Target argmax recall and training-query argmax agreement remain $100.0 \pm 0.0\%$ across five seeds.

Sequential Edits Inserted	1	5	10	20	30	40
Total Active Pool Size	161	165	170	180	190	200
Cumulative Efficacy (All Past Edits)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Cumulative Generality (Paraphrases)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Background Specificity (Original 160)	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
Mean Attention Mass $\mu_x(\omega^*)$	99.84%	99.76%	99.46%	99.60%	99.66%	99.70%

313 6.5. Lifelong Cumulative Knowledge Editing Benchmark

314 To evaluate whether the atom-space formulation provides lifelong memory sta-
 315 bility, we conduct a cumulative insertion experiment:

- 316 1. We train the base model on 160 baseline atoms (initial accuracy: 100.0%).
- 317 2. We insert all 40 held-out technical facts *sequentially and permanently without*
 318 *teardown*, expanding the active pool from 161 to 200 atoms.
- 319 3. At regular intervals, we evaluate Cumulative Efficacy across all facts in-
 320 serted up to that point, Cumulative Generality across paraphrases, Background
 321 Specificity across the original 160 training facts, and the Mean Attention Mass
 322 $\mu_x(\omega^*)$ allocated to the target atoms.

323 Table 5 and Figure 1 present the results of this lifelong evaluation.

324 Across five seeds, all 39 earlier insertions remain correctly classified when the
 325 40th record is added, and no argmax changes are observed on the 160 training
 326 queries. At 40 insertions, mean target-atom attention mass is $99.681 \pm 0.014\%$.
 327 These observations apply to this small exact-scoring benchmark; the truncation
 328 theorem does not itself prove resistance to forgetting or memory dilution.

329 6.6. Dynamic Tri-Space Routing and Path Ablation

330 To evaluate the integrated tri-space architecture of Section 3, we construct an
 331 empirical environment with three task domains assigned to different information
 332 paths:

- 333 1. **In-Context Ephemeral Reasoning (Ω_{ctx}):** Input sequences provide ad-hoc
 334 premises unavailable in world memory (e.g., synthetic cryptographic bind-
 335 ings). The model must attend to local token positions Ω_{ctx} .

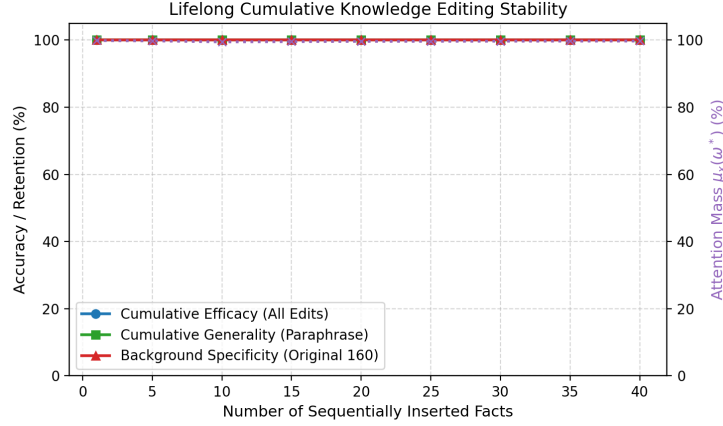


Figure 1: Lifelong cumulative knowledge editing stability across 40 sequential fact insertions. Cumulative efficacy, generality, and background specificity remain perfectly stable at 100%, while the query-conditioned attention probability mass $\mu_x(\omega^*)$ remains tightly concentrated at $> 99.6\%$ despite continuous memory expansion.

2. **Corpus World Knowledge (Ω_{corpus}):** Prompts ask declarative technical questions from our factual repository without providing premise context. The model must route to Ω_{corpus} .
3. **Parameter-Resident Expert Outputs (Ω_{params}):** Queries describe small procedural classification tasks. Trigger keys select trainable, unit-normalized expert vectors stored as model parameters rather than corpus values.

The tri-space router $\lambda(x) \in \Delta^2$ is parameterized as a lightweight projection network trained jointly with the representation heads. To prevent the norm scale mismatch analyzed in Section 3, parameter-expert keys are normalized to the unit sphere, matching the geometric scale of the corpus embeddings. Evaluation uses unseen query strings with the identical wrapper “Determine the output for:” across all three families, removing explicit route words such as “premise,” “stored,” and “expert.”

Table 6 and Figure 2 present the surface-matched results:

- **Routing Generalization:** Mean route accuracy is 100.0%, 82.4%, and 100.0% for context, corpus, and parameter tasks. Mean intended-space allocation is 68.0%, 56.6%, and 93.7%, respectively.
- **Task and Ablation Results:** Mean full-model accuracy is 100.0%, 96.8%, and

Table 6: Five-seed surface-matched routing and ablation (mean $\pm$ sample standard deviation). Route accuracy is the fraction whose largest gate selects the intended space. Tasks and labels are known during training, but evaluation strings are unseen and share one wrapper across families.

Task Domain	Task Acc.	Route Acc.	Mean Gating Distribution $\lambda(x)$			Ablated Acc.
			λ_{ctx}	λ_{corpus}	λ_{params}	
In-Context Reasoning (Ω_{ctx})	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	68.0$\pm$2.1	6.2 $\pm$ 0.9	25.8 $\pm$ 1.3	0.0 $\pm$ 0.0
Corpus World Knowledge (Ω_{corpus})	96.8 $\pm$ 1.8	82.4 $\pm$ 0.9	12.9 $\pm$ 0.7	56.6$\pm$2.2	30.5 $\pm$ 1.7	0.0 $\pm$ 0.0
Parameter Expert Outputs (Ω_{params})	100.0 $\pm$ 0.0	100.0 $\pm$ 0.0	3.1 $\pm$ 0.2	3.1 $\pm$ 0.4	93.7$\pm$0.5	0.0 $\pm$ 0.0

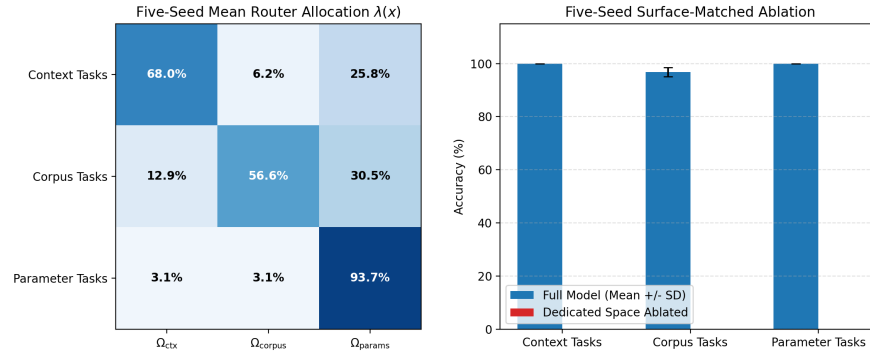


Figure 2: Five-seed surface-matched tri-space evaluation. Left: mean router allocation on unseen strings with a shared wrapper. Right: mean task accuracy with sample-standard-deviation error bars; intended-path ablation reduces accuracy to zero.

354 100.0%; ablating the intended space reduces every corresponding accuracy to
355 0% in all five runs.

356 Removing explicit route words exposes partial corpus confusion with the param-
357 eter path. These results establish limited surface-form robustness over the same
358 records, operations, and labels. They do not establish transfer to held-out task
359 families, expert composition, or adversarial robustness.

360 6.7. Knowledge Contradiction and Anti-Atom External-Memory Suppression

361 In practical applications, external-memory records may require correction or
362 suppression. We examine both operations in a controlled signed-read classifier; we
363 do not evaluate physical deletion, privacy guarantees, or removal from pretrained
364 parameters.

365 *External-Memory Suppression via Anti-Atoms.* To test zero-retraining suppres-
366 sion in the external read, five synthetic sensitive records are inserted into the base

Table 7: External-memory suppression and counterfactual updating via signed reads, using a toy 49-token output vocabulary. Exact matched pairs reduce target probability to approximately $1/|\mathcal{V}| = 2.04\%$ in this external-read-only classifier. Clean-control argmax accuracy remains 100.0%.

Target Entity / Query	Pre-Edit P(Target)	Post-Edit P(Target)	Status	Background Specificity
<i>External-Memory Suppression via Anti-Atoms (unit mass under v^-)</i>				
Secret Base Alpha Coordinates	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
Patient Confidential Diagnosis	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
Proprietary Patent Number	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
Classified Document Location	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
VIP Cryptographic Key	99.998%	2.04% ($1/ \mathcal{V} $)	Suppressed	100.0%
<i>Counterfactual Competitive Updating ($w_{\text{new}} = 2.5$, Append-Only)</i>				
Ethereum $\rightarrow$ Proof-of-Stake	3.0% (old)	40.0% (new)	Overridden	100.0%
Java $\rightarrow$ Oracle	0.01% (old)	60.1% (new)	Overridden	100.0%
Pluto $\rightarrow$ Dwarf Planet	32.3% (old)	23.9% (new)	Competing	100.0%
Twitter $\rightarrow$ X Corp	99.8% (old)	0.04% (new)	Lexical Bias	100.0%
Python $\rightarrow$ PyPy JIT	98.9% (old)	0.24% (new)	Lexical Bias	100.0%

corpus. Without gradient descent or index reconstruction, we insert an anti-atom ω^- with positive mass 1.0 under v^- and value $v^- = E[y_{\text{sensitive}}]$ at the same metric coordinate as each record. This benchmark evaluates matched-query output suppression, not certified deletion from pretrained weights.

Table 7 and Figure 3 (right) show that matched anti-atoms reduce target-token probability to approximately **2.04%**. In a vocabulary of $|V| = 49$ tokens, $1/49 \approx 0.0204$. The paired contribution is exactly zero by construction; small background contributions remain in the complete read. Argmax accuracy on 35 clean control facts remains **100.0%**.

Approximate cancellation stress test. We next perturb the anti-atom key, value, mass, or query over 50 random trials at magnitudes 0, 0.01, 0.05, 0.10, and 0.20, with background sizes 10, 100, and 1000. The analytical bound covers all 3000 trials. At $N = 100$, key perturbations of 0.01, 0.05, and 0.10 leave mean pair residuals of 4.5%, 72.1%, and 98.0% relative to the original positive contribution; corresponding value perturbations leave 5.5%, 25.5%, and 40.5%, while 1%, 5%, and 10% mass errors leave 0.5%, 2.4%, and 4.8%. Query perturbation alone leaves numerical-zero residual because both exactly matched keys receive the same changed kernel weight. Exact anti-atoms are therefore algebraically effective but fragile to key mismatch.

Competitive Updating and the Lexical Bias Phenomenon. In an append-only log where old atoms cannot be physically deleted, we evaluate updating five technical

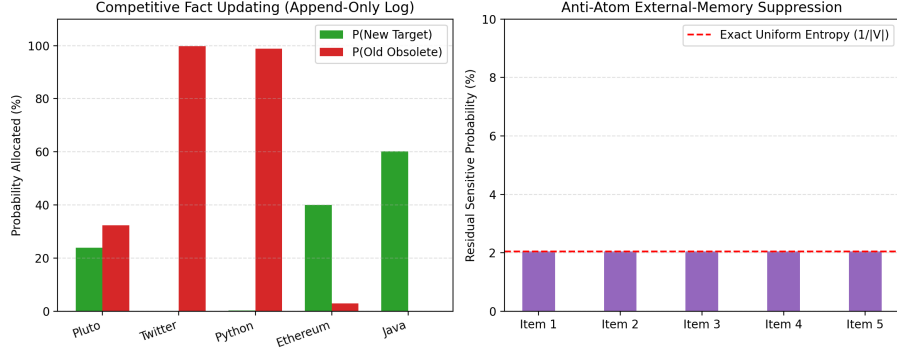


Figure 3: Empirical evaluation of signed atom spaces. Left: Competitive counterfactual updating in an append-only log under passive recency weighting ($w_{\text{new}} = 2.5$). Right: matched anti-atom cancellation drives residual target probability to uniform entropy ($P = 2.04\% \approx 1/|\mathcal{V}|$) in the external read.

statements by inserting contradictory atoms ω_{new} with priority recency weighting $w_{\text{new}} = 2.5$. Across five seeds, competitive updating succeeds on $44.0 \pm 8.9\%$ of the five records per run; the canonical seed succeeds for two records (*Ethereum* $\rightarrow$ *Proof-of-Stake* and *Java* $\rightarrow$ *Oracle*). In the remaining canonical cases, similarity to the old key outweighs the fixed recency weight under the exponential kernel. Matched anti-atom suppression succeeds on $100.0 \pm 0.0\%$ of records, with mean residual probability $2.0407 \pm 0.00002\%$ and $100.0 \pm 0.0\%$ background specificity.

This small result shows that a fixed passive recency weight is not sufficient for every evaluated update. Active cancellation is one possible design response, but broader benchmarks are required before selecting an update policy.

6.8. Autoregressive Generative Modeling with GPT-2

To establish that the atom-space read operator functions effectively in multi-layer causal generative architectures, we hook Ω_{corpus} directly into the final hidden representation of a pretrained transformer (gpt2, 124M parameters [16]).

Continuous Hidden-State Augmentation with Relevance Gating. At each greedy decoding step t , the current textual prefix $x_{1:t}$ is encoded by the frozen sentence encoder, $q_t = \phi_q(x_{1:t}) \in \mathbb{R}^m$, and matched against the corpus atom space. GPT-2 independently maps the token prefix to $h_t \in \mathbb{R}^{d_{\text{model}}}$. A matching atom is injected only if its cosine score exceeds 0.60 and it has not already been consumed in the current generation. This one-shot rule prevents repeated injection of the same

Table 8: Selected rows from a ten-fact greedy GPT-2 (124M) evaluation. Factual atoms boost first-token target probabilities and flip all ten canonical first predictions.

Factual Target Task	Target Probability $P(y^*)$		Ratio	Top Predicted Token	
	Base GPT-2	Atom-Space		Base GPT-2	Atom-Space
QUIC Transport Protocol (UDP)	2.64%	70.00%	26.5×	the	UDP
Milky Way Black Hole (Sagittarius)	0.47%	64.33%	136.3×	after	Sag
Safe Systems Language (Rust)	0.03%	32.99%	1116.0×	a	Rust
Uranium Radiative Decay (lead)	0.87%	9.80%	11.3×	uranium	lead
CRISPR Bacterial Defense (viruses)	2.25%	38.41%	17.1×	infect.	viruses

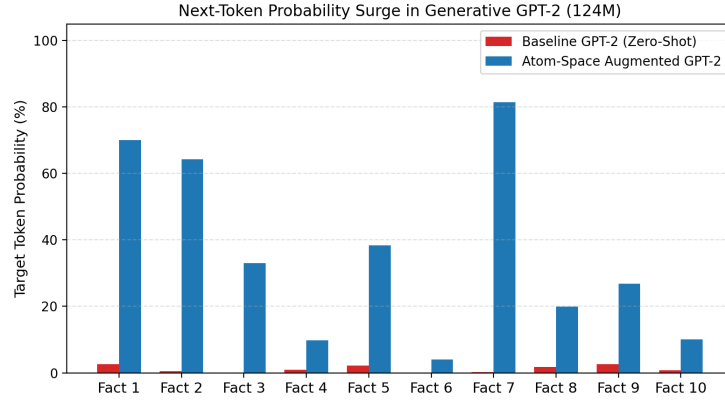


Figure 4: First-token target probabilities in GPT-2 (124M) before and after direct target-embedding intervention. The argmax changes toward the desired first token or subtoken on all five selected prompts.

408 target token:

$$\tilde{h}_t = \begin{cases} (1 - \lambda_{\text{read}})h_t + \lambda_{\text{read}}W_{\text{proj}}z_t, & \text{relevant and unconsumed,} \\ h_t, & \text{otherwise.} \end{cases}$$

409 feeding directly into the language model head: $p(x_{t+1} | x_{1:t}) = \text{softmax}(W_{\text{LM}}\tilde{h}_t)$.
 410 Each atom’s value v_i is mapped to the token embedding of the target entity from
 411 GPT-2’s vocabulary matrix.

412 *First-Token Correction and Multi-Token Continuation.* Table 8 and Figure 4 sum-
 413 marize the quantitative performance. Base GPT-2 assigns $< 3\%$ probability to
 414 the correct factual token in all cases, generating severe hallucinations upon free
 415 autoregressive decoding:

- 416 • **Milky Way Black Hole Prompt:** Base GPT-2 generates: “*The supermas-*
417 *sive black hole at the center of the Milky Way is named **after the famous as-***
418 ***tronomer Johannes Kepler, who discovered the black hole...***”. The interven-
419 tion changes the first generated subtoken from after to Sag (0.47% $\rightarrow$ 64.33%
420 target probability), and the tokenizer completes “Sagittarius A” before the con-
421 tinuation later becomes unreliable.
- 422 • **Uranium Decay Prompt:** Base GPT-2 generates “*natural uranium decays*
423 *radiatively into stable isotopes of **uranium...***”. The intervention changes the
424 first generated token to lead (0.87% $\rightarrow$ 9.80%), but the later phrase “lead,
425 thorium, and lead-based materials” is not a reliable scientific continuation.
- 426 • **Systems Language Prompt:** Base GPT-2 generates generic filler: “*The sys-*
427 *tems programming language offering memory safety without garbage collec-*
428 *tion is **a great way to get around the limitations...***” Augmented with the atom
429 space, probability surges by 1116 $\times$ (0.03% $\rightarrow$ 32.99%), generating: “*The sys-*
430 *tems programming language offering memory safety without garbage collec-*
431 *tion is **Rust. The Rust compiler is a powerful tool for...***”

432 Across all ten canonical prompts, the intervention changes the first-token argmax
433 to the desired token or subtoken. Teacher-forced scoring over complete target
434 strings obtains 96.7% target-token accuracy, reflecting one later-subtoken error for
435 “chloroplast.” On ten manually constructed paraphrases, retrieval selects the in-
436 tended atom and changes the first-token argmax on 9/10. Across ten unrelated
437 control texts, the similarity threshold prevents activation at every teacher-forced
438 step, so the augmented/base perplexity ratio is exactly 1.0 by construction. A grid
439 over thresholds $\{0.4, 0.5, 0.6, 0.7, 0.8\}$ and $\lambda_{\text{read}} \in \{0.25, 0.5, 0.75, 0.85\}$ records
440 factual and false activation separately. These tests remain too small to establish
441 naturalness, broad language preservation, or factuality gains.

442 *Naive Full-Parameter Fine-Tuning Baseline.* To give the atom-space read a first
443 parametric point of comparison, we fine-tune all of GPT-2’s parameters on each
444 of the ten canonical facts in isolation: starting from the pretrained checkpoint, we
445 take gradient steps (AdamW, learning rate 10^{-5} , up to 25 steps, early-stopped at
446 training loss < 0.01) on the canonical prompt-target pair, then reset to the clean
447 checkpoint before the next fact. This is the standard unconstrained “FT” baseline
448 from the model-editing literature.

449 *ROME-Lite: A Simplified Locate-and-Edit Baseline.* We additionally implement
450 a simplified, single-layer rank-one edit in the style of ROME [3] on the same

Table 9: Naive full-parameter fine-tuning and a simplified single-layer rank-one edit (ROME-lite) versus the relevance-gated atom-space read on the same ten GPT-2 facts, paraphrases, and control texts. Neighborhood specificity has no atom-space analogue because the read injects no weight changes.

Metric	Atom-Space Read	Naive Fine-Tuning (FT)	ROME-Lite
Canonical efficacy (first-token argmax)	100% (10/10)	100% (10/10)	100% (10/10)
Paraphrase generality (first-token argmax)	90% (9/10)	70% (7/10)	40% (4/10)
Neighborhood specificity (other 9 facts' argmax unchanged)	n/a (no weight change)	58.9% (mean)	87.8% (mean)
Control-text perplexity ratio (locality)	1.000 (exact, by construction)	0.982 mean, 1.022 max	0.996 mean, 1.004 max
Mechanism	Relevance-gated read, zero weight updates	Full-parameter gradient descent, mean 18.0 steps/fact	Single-layer rank-one edit, identity-covariance closed form

ten facts, so all three methods share identical prompts, paraphrases, neighborhood facts, and control texts. For each fact we (i) select an edit layer among $\{2, 4, 6, 8, 10\}$ via a lightweight single-token causal trace (Gaussian corruption of the final prompt token’s embedding, then per-layer MLP restoration, keeping the layer with the largest probability recovery); (ii) take the key k as the clean model’s MLP down-projection input at that token and optimize a value v^* with teacher-forced cross-entropy only (20 Adam steps, no KL “essence” regularizer); and (iii) insert the association with the closed-form identity-covariance rank-one update $W \leftarrow W + k(v^* - k^\top W)^\top / (k^\top k)$ to the down-projection weight. This omits ROME’s multi-template key averaging, precomputed key-covariance statistics, and KL regularization against a separate prompt set, and is evaluated at the same ten-fact scale as the fine-tuning baseline; it is a same-scale, same-metric comparison point, not a reproduction of ROME’s published numbers.

Table 9 shows that naive fine-tuning matches the atom-space read’s canonical efficacy but generalizes less well to paraphrases (70% vs. 90%) and, more importantly, changes the argmax on 41.1% of the other nine facts’ canonical queries on average even though each fact is edited in isolation and training loss converges below 0.01. Mean control-text perplexity drifts by up to 2.2% in the worst case, whereas the atom-space read leaves unrelated text perplexity exactly unchanged because unrelated prompts never cross the activation threshold. ROME-lite’s single-layer, identity-covariance edit reduces neighborhood argmax drift substantially relative to full-parameter fine-tuning (12.2% vs. 41.1% of other facts changed) and keeps control-text perplexity closer to the pre-edit model (0.4% mean drift vs. 1.8%), consistent with the locality benefits reported for locate-and-edit methods in the literature. However, at this simplified, single-template scale it generalizes to paraphrases substantially worse than either the fine-tuning baseline or the atom-space read (40% vs. 70% and 90%), plausibly because omitting ROME’s multi-template key averaging leaves the inserted association keyed too tightly to the single canonical prompt’s representation. This three-way,

480 same-scale comparison illustrates both the catastrophic-interference risk that mo-
 481 tivates locate-and-edit methods and the specificity/generalizability trade-off between
 482 them, but remains restricted to ten facts on GPT-2 (124M) with a simplified,
 483 non-covariance-regularized edit; a full-fidelity reproduction of ROME, MEMIT,
 484 WISE, or AlphaEdit at CounterFact scale [3, 4, 14, 15] remains future work.

485 7. Discussion and Related Work

486 The individual components unified under our framework have extensive histo-
 487 ries in machine learning:

- 488 • **Retrieval and Memory Networks:** k NN-LM [6], RETRO [7], and Memory
 489 Networks [17, 12] correspond directly to the corpus read $R(x; \Omega_{\text{corpus}})$.
- 490 • **Parameter Routing and Mixture-of-Experts:** MoE models [8, 18, 10] and
 491 dynamic LoRA routing [9] represent realizations of $R(x; \Omega_{\text{params}})$, where expert
 492 selection reflects a discrete, sparse measure over parameter patches.
- 493 • **In-Context Attention:** Standard multi-head attention [1] is an integral
 494 $R(x; \Omega_{\text{ctx}})$ over discrete sequence indices.

495 What this formulation contributes beyond existing empirical literature is five-
 496 fold:

- 497 1. **A Common Mathematical Object:** By expressing context, corpus retrieval,
 498 and parameter routing as integrals over measurable atom spaces, these mecha-
 499 nisms can be analyzed within a unified framework. Surface-matched evaluation
 500 quantifies both successful routing and corpus-to-parameter confusion.
- 501 2. **A Checkable Retrieval Condition:** The Lipschitz-margin inequality gives a
 502 sufficient condition for exact top- k inclusion. It is not by itself a guarantee of
 503 edit success, generalization, or specificity.
- 504 3. **A Quadrature Foundation for Vector Retrieval:** Theorem 1 and Corollary 1
 505 link the read integral to exact ranked truncation and error-budgeted k . HNSW
 506 candidate recall is measured separately [19].
- 507 4. **Cumulative External-Memory Insertion:** External records are additive over
 508 the measured atom space ($\Omega_t = \Omega_0 \cup \{\omega^*\}$). In the 200-record experiment, no
 509 argmax changes are observed on training controls. Section 6.8 adds two para-
 510 metric comparison points at matched scale: a naive full-parameter fine-tuning

511 baseline that matches canonical efficacy but shows weaker paraphrase gener-
512 ality and substantial neighborhood drift, and a simplified single-layer rank-
513 one edit (ROME-lite) that recovers most of that lost neighborhood specificity
514 while generalizing to paraphrases worse than either alternative. A full-fidelity
515 comparison against ROME [3], MEMIT [4], and recent lifelong or null-space-
516 constrained editors [14, 15] at CounterFact scale is left for future work.

517 **5. Zero-Retraining External-Memory Suppression:** Signed reads enable
518 matched anti-atoms to cancel selected external-memory contributions without
519 weight retraining or index reconstruction.

520 Finally, our generative experiment applies relevance-gated atom reads through-
521 out greedy decoding, demonstrating first-token factual steering without fine-
522 tuning.

523 7.1. Limitations

524 The trainable synthetic studies use five seeds but remain small and templated.
525 Frozen exact 1-nearest-neighbor retrieval matches the trained read on synthetic in-
526 sertions. CounterFact retrieval uses only 500 records and does not compare end-to-
527 end generation or parameter editing against ROME, MEMIT, MEND, WISE, Al-
528 phaEdit, or RAG [3, 4, 14, 15]; the naive full-parameter fine-tuning and simplified
529 single-layer rank-one (ROME-lite) baselines in Section 6.8 are first, restricted com-
530 parison points on only ten facts on GPT-2 (124M), and the latter omits ROME’s
531 multi-template key averaging, precomputed key-covariance statistics, and KL reg-
532 ularization, so it does not reproduce ROME’s published numbers. Exact adaptive
533 certification computes all pairwise scores; its large selected fractions show that it is
534 not itself a scaling solution. The HNSW corpus is too small for large-scale latency
535 conclusions. Tri-space tasks and labels are known during training, and parameter
536 experts are output vectors rather than subnetworks or LoRA patches. Anti-atoms
537 retain both records and do not remove pretrained knowledge; key mismatch rapidly
538 weakens cancellation, and extraction/privacy attacks are not evaluated. The GPT-
539 2 study has ten facts, ten paraphrases, and ten controls, so it remains a mechanism
540 study rather than evidence of general hallucination reduction.

541 8. Conclusion

542 We have presented a measure-theoretic abstraction that casts finite attention,
543 dense retrieval, and parameter-expert output routing as weighted reads into a com-
544 mon output space. We derived a sufficient condition for top- k inclusion, exact-

545 ranked truncation bounds, an adaptive error-budget rule, and an approximate can-
546 cellation bound, then examined them in controlled synthetic and CounterFact stud-
547 ies.

548 The synthetic insertion studies obtain perfect five-seed target classification and
549 retention, but frozen exact 1-nearest-neighbor retrieval obtains the same result.
550 Error-budgeted exact truncation achieves complete empirical coverage while often
551 requiring a large fraction of the corpus. CounterFact reveals strong exact retrieval
552 but weak fixed-threshold paraphrase activation and nonzero neighborhood activa-
553 tion; HNSW recovers the exact top-1 among ten candidates in this small run. The
554 remaining studies show imperfect routing, fragile approximate cancellation, and
555 limited GPT-2 factual-token steering. Competitive end-to-end baselines, larger
556 corpora and models, held-out task families, privacy attacks, and broad language
557 evaluations remain necessary before comparative performance claims.

558 **CRedit Authorship Contribution Statement**

559 **Farshid Mossaiby:** Conceptualization, Methodology, Formal analysis, Soft-
560 ware, Investigation, Validation, Data curation, Writing - original draft, Writing -
561 review & editing.

562 **Declaration of Competing Interest**

563 The author declares that they have no known competing financial interests or
564 personal relationships that could have appeared to influence the work reported in
565 this paper.

566 **Data Availability**

567 The complete open-source codebase, synthetic and technical benchmarks, ex-
568 perimental suites, and reproduction scripts generated during this study are publicly
569 available in the GitHub repository at <https://github.com/mossaiby/RoAS>.

570 **Declaration of Generative AI in Scientific Writing**

571 During the preparation of this work, the author utilized Google Gemini to as-
572 sist in the mathematical formalization of quadrature bounds, experimental pipeline
573 design, and manuscript formatting. After using this tool, the author reviewed and
574 edited the content as needed and takes full responsibility for the content of the
575 publication.

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